min $z = x_1 + x_2 + x_3$

$x_1 + x_2 \geq x_1 + x_3 \geq x_2 + x_3 \geq 1 \quad x_1, x_2, x_3 \in \{0, 1\}$

$-x_1 - x_2 \leq -1 \quad -x_1 - x_3 \leq -1 \quad -x_2 - x_3 \leq -1$

$Ax \leq b$

$$\begin{pmatrix} -1 & -1 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \leq \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}$$

(0,0,0) - infeasible (no conditions met)

(1,0,0) - infeasible (fails $-x_2 - x_3 \leq -1$)

(0,1,0) - infeasible (fails $-x_1 - x_3 \leq -1$)

(0,0,1) - infeasible (fails $-x - x_2 \leq -1$)

(1,1,0) - feasible, $z = 2$

(0,1,1) - feasible, $z = 2$

(1,1,1) - feasible, $z = 3$

minimum value $z^* = 2$

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